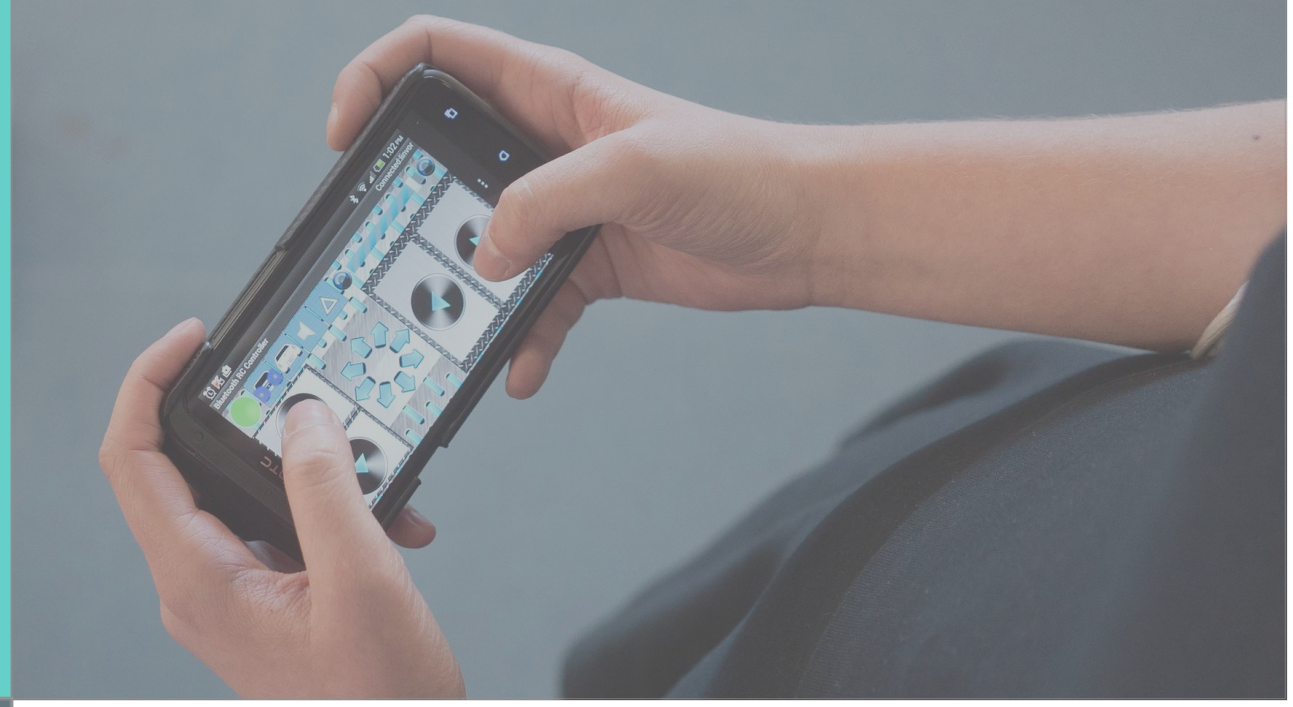
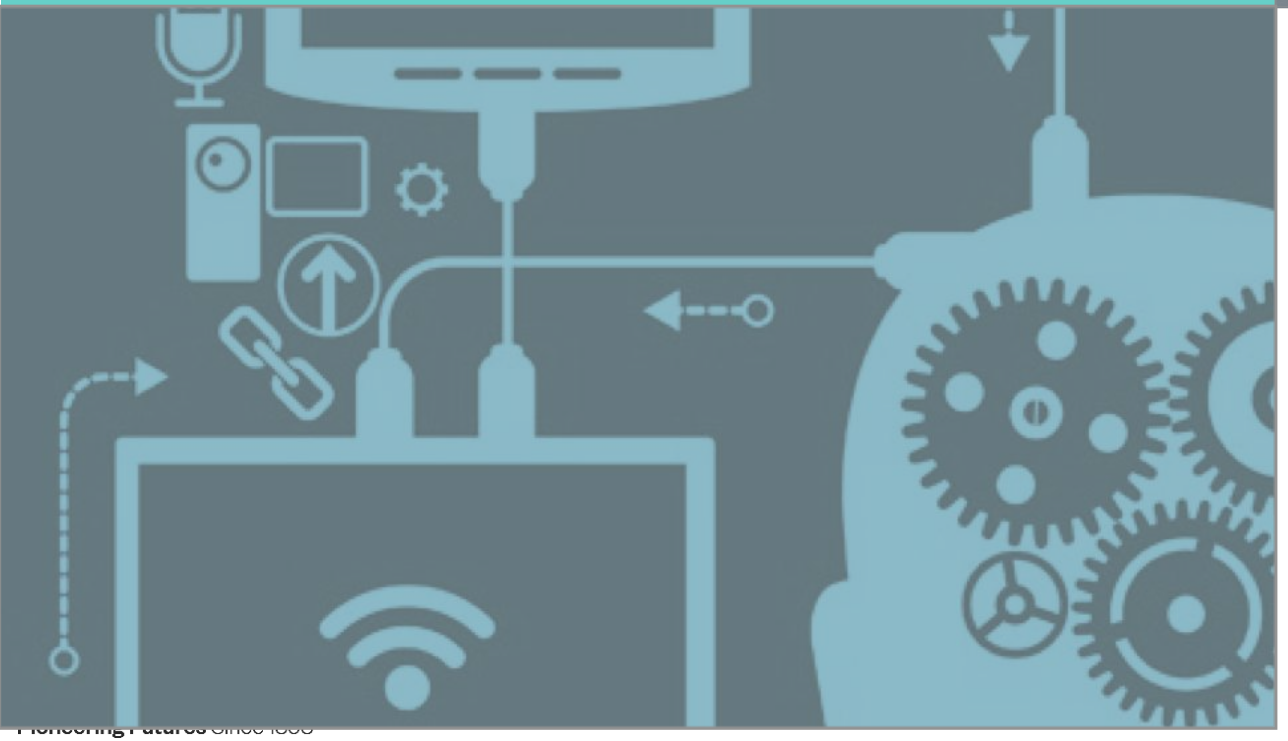
CN6000
Final Year Project 2025/26

Topic 9
Quantitative Research Methods





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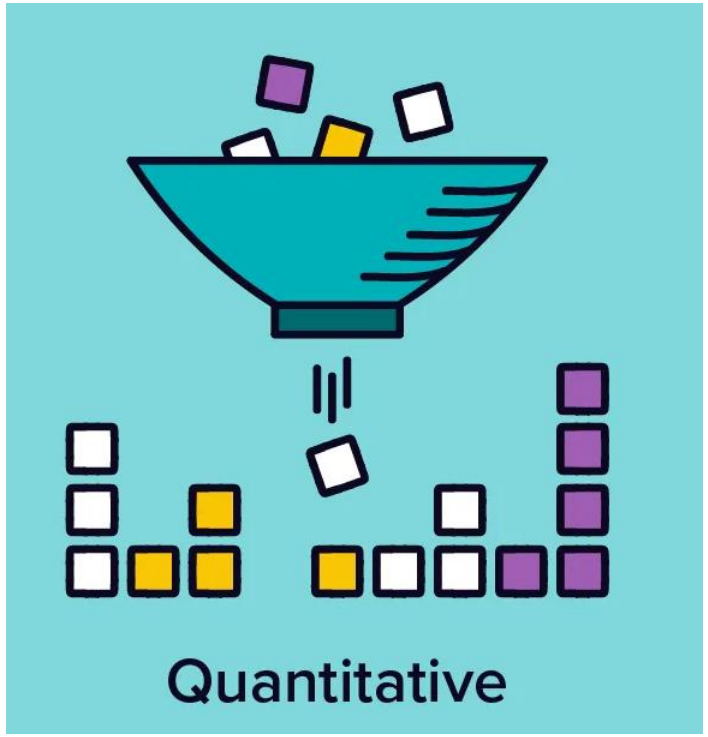
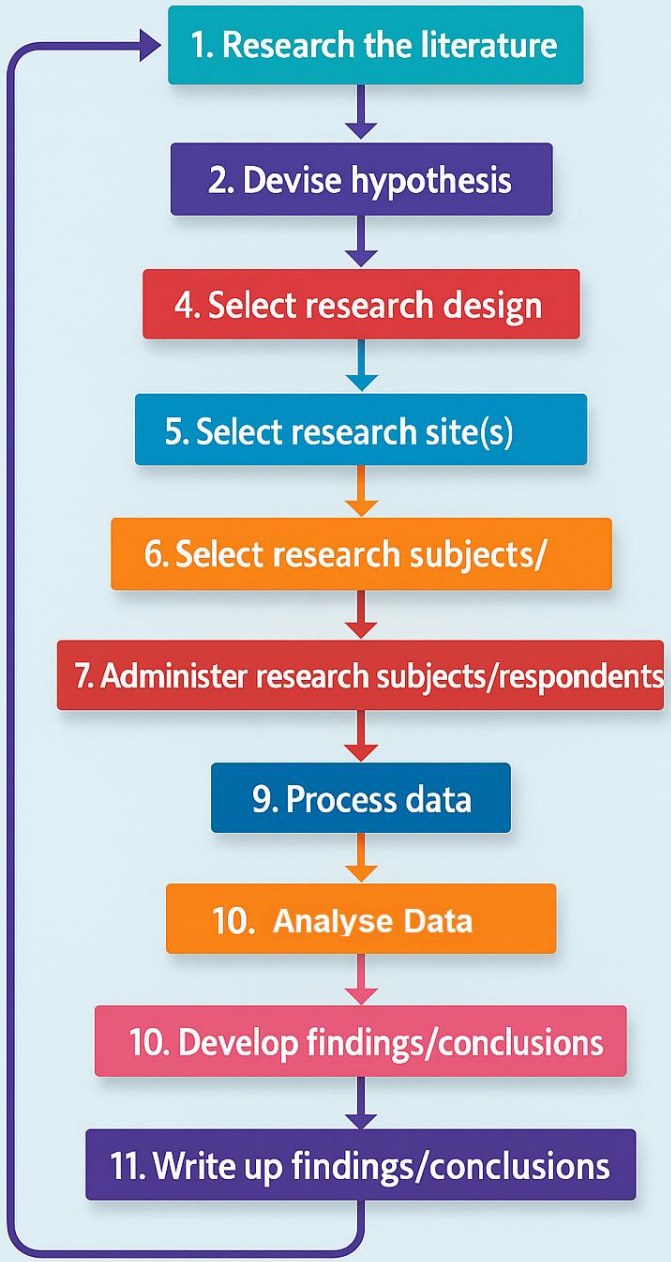
Pioneering Futures Since 1898

What is Quantitative Research?

- Model and measure variables for research purposes.
- Identify data sources, execute research and argue about the findings
- Answer questions by testing **hypotheses** (deductive approach) and **inferencing** (derive by reasoning) **conclusions.**



The process of quantitative research



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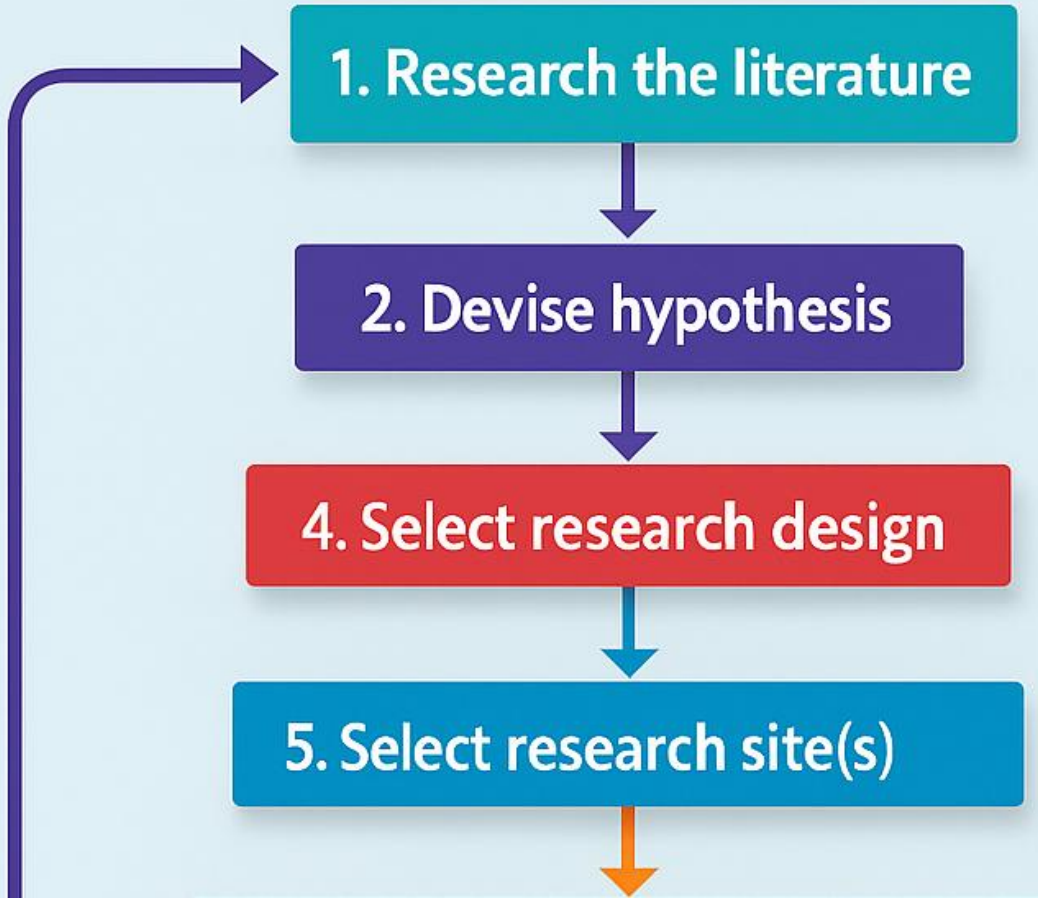
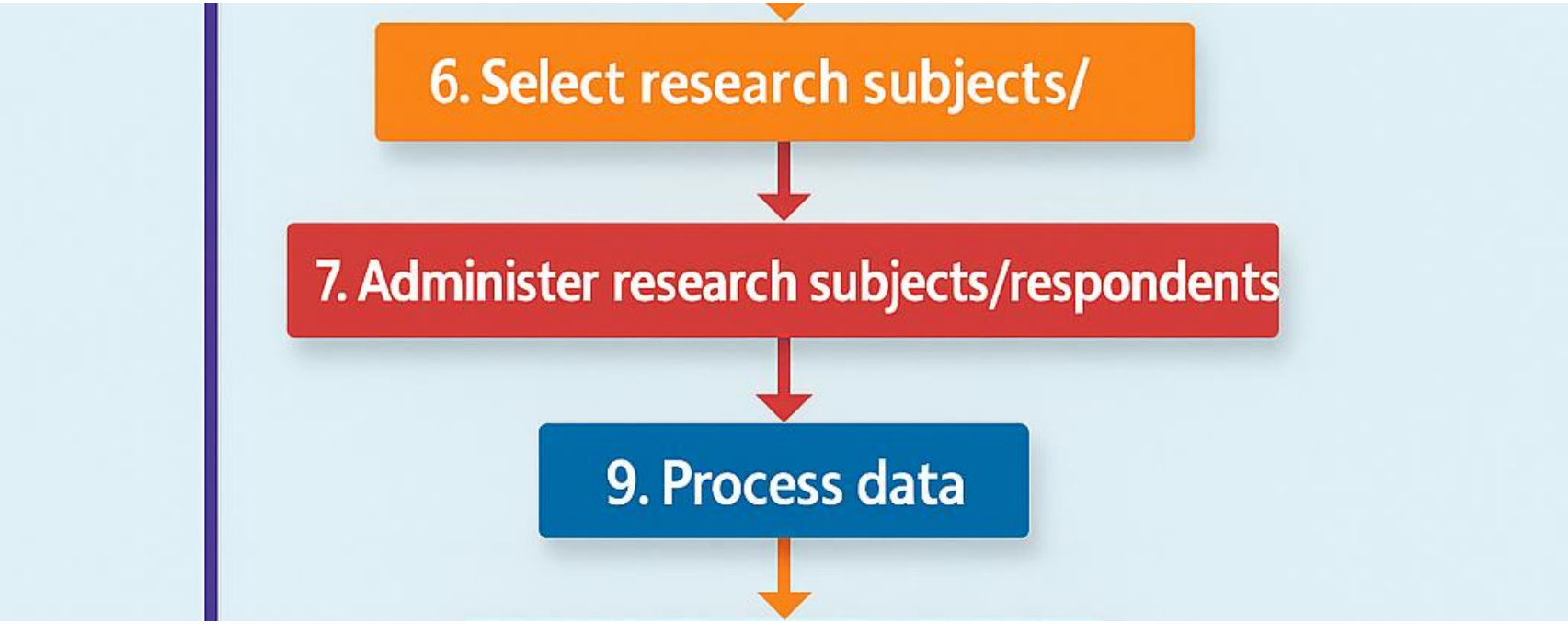
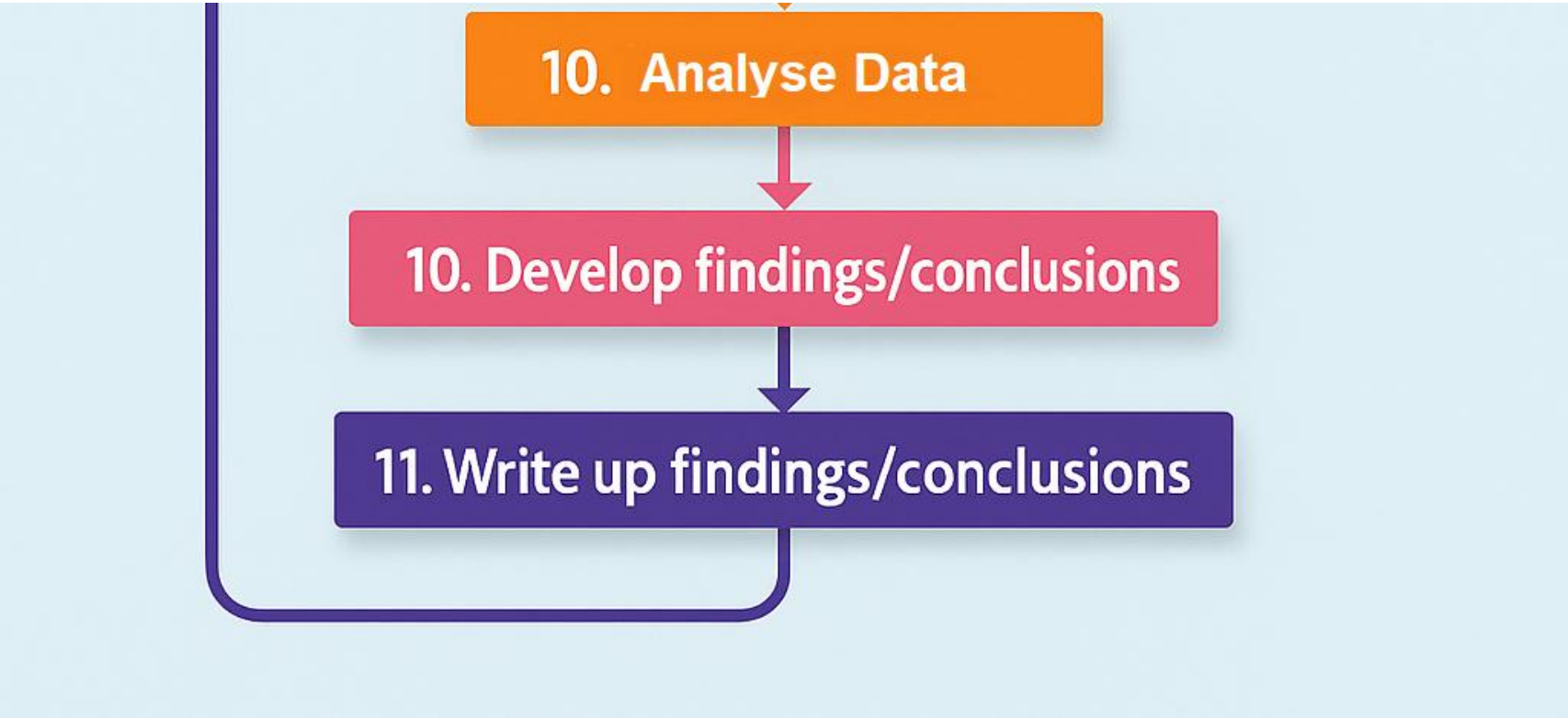
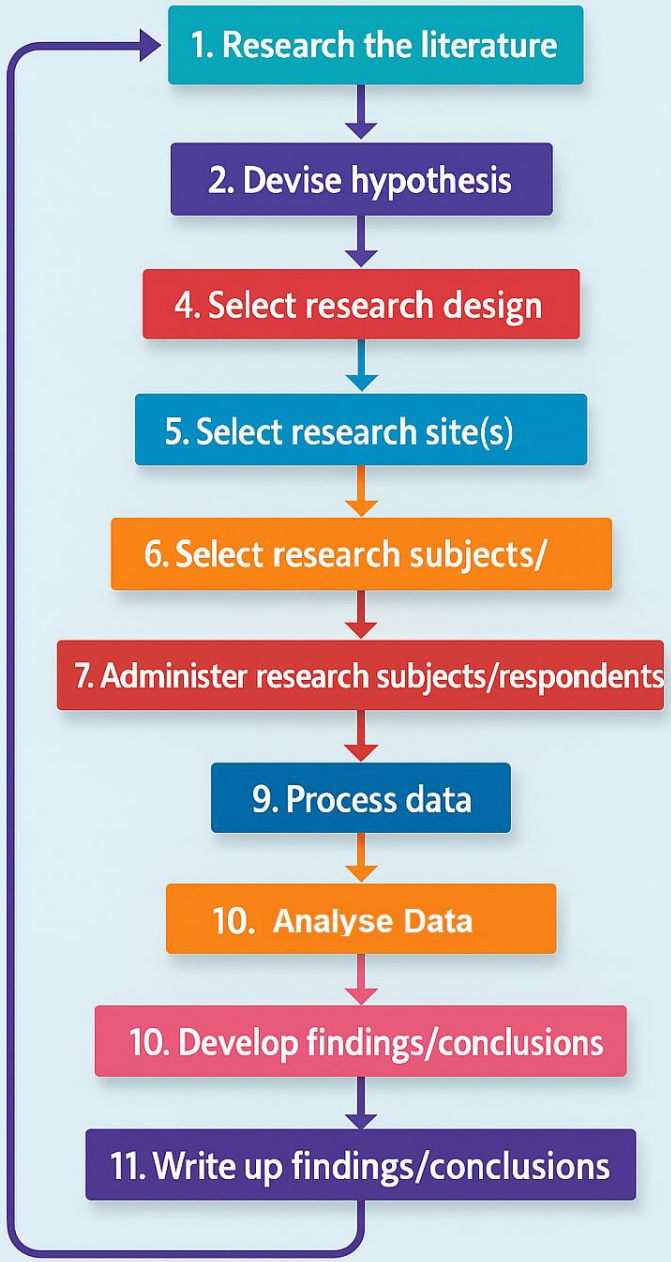



6. Select research subjects/
7. Administer research subjects/respondents
9. Process data


10. Analyse Data
10. Develop findings/conclusions
11. Write up findings/conclusions

The process of quantitative research



Hypothesis

- **Prediction of what the research will find**
- **Empirical statement-verified based upon observation, experiment, survey, etc**
- **Testable to be true or false through the research findings**

Example: **Time management app will positively affect employees' job performance.**

Do not forget to state your null hypothesis!

Hypothesis

- **What is Null Hypothesis?**
- Denoted with H_0 is that two possibilities are the same.
- **It is all the other possibilities and by trying to prove the null and 'Failing' you can then prove your hypothesis.**
- Sometimes called 'Alternative Hypothesis' and should be mutually exclusive – meaning no other possibilities.

Example: H_1:**Time management app will positively affect employees' job performance.**

H_0: **Time management app will negatively or not affect employees' job performance.**

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Directional vs. non-directional hypotheses

Non-directional hypotheses are those that do postulate a relationship or difference but offer no indication of the direction of these relationships or differences.

Example

Hypothesis: H_1: There is a relationship between age and willingness to use AI.

What would be the null hypothesis here?

Directional vs. non-directional hypotheses

Non-directional hypotheses are those that do postulate a relationship or difference, but offer no indication of the direction of these relationships or differences.

Example

Hypothesis: H_1: There is a relationship between age and willingness to use AI.

What would be the null hypothesis here?

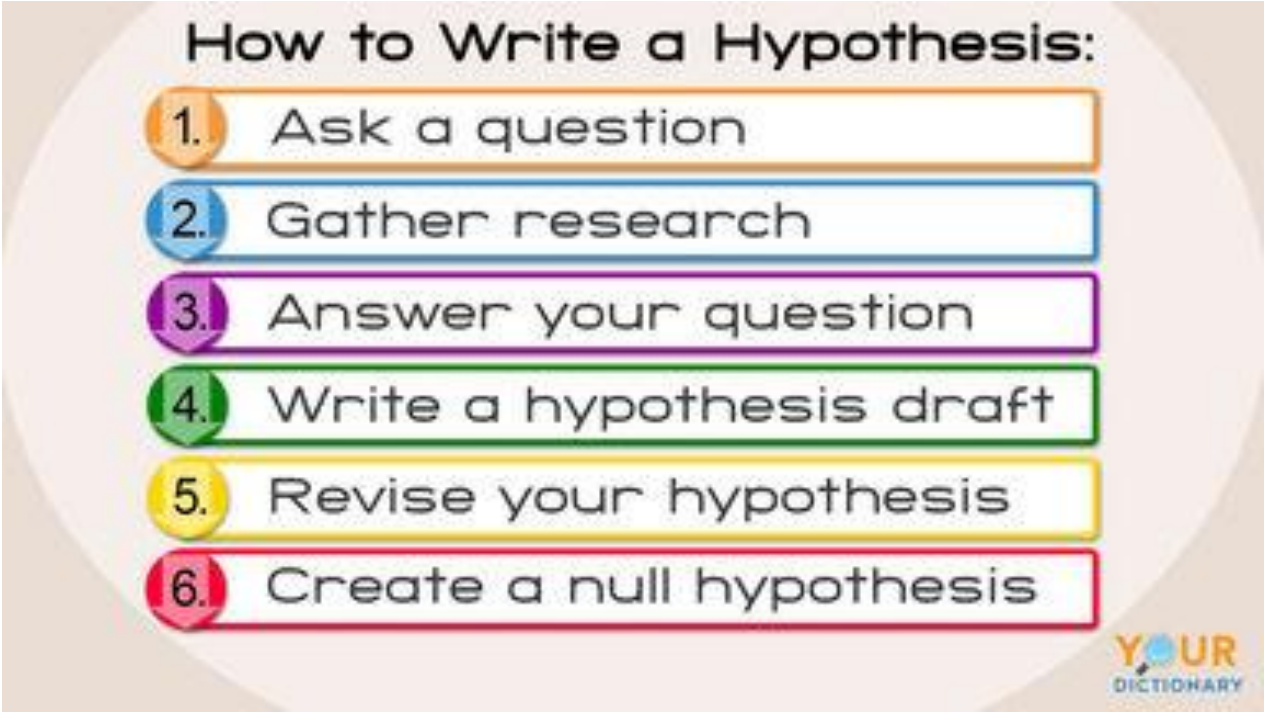
H_0: There is no relationship between age and willingness to use AI.

Directional vs. non-directional hypotheses

Directional hypotheses are preferred in scientific research!

Common mistakes

- Students not providing hypotheses
- The hypotheses are not the correct format!



https://grammar.yourdictionary.com/for-students-and-parents/how-create-hypothesis.html

Have a think:

If we want to collect quantitative data, what type of questions do we need to ask?

Well actually, before we decided the questions, we need to see what type of data we can use in Quantitative research.

We said we care about number and we do our analysis to generate number….so here we go.

How do we collect data:

- **Interval or ratio**: regular distances between all categories in range (e.g. Age group) or an exact number (e.g. speed).
- **Ordinal**: categories can be ranked, but unequal distances between them (e.g. Likert scale 1 to 5)
- **Nominal**: qualitatively different categories that cannot be ranked (e.g. colour of eyes)
- **Dichotomous or binary**: only two categories (e.g. gender)

Example of Ratio

Data set

Person	Weight (kg)
1	70
2	55
3	90
4	75
5	66
6	73
7	80
8	58
9	76
10	81

Descriptive statistics

Statistic	Result	Formula
Mean (average)	72.4	=AVERAGE
Median	74	=MEDIAN
Standard Deviation	10.637	=STDEV.S
Variance	113.15556	=VAR.S
Skewness	-0.223701	=SKEW

Types of variables

Ratio

Ratio scales are the ultimate data type when it comes to measurement scales because they tell us about the order, they tell us the exact value between units, AND they also have an absolute zero–which allows for a wide range of both descriptive and inferential statistics to be applied. At the risk of repeating myself, everything above about interval data applies to ratio scales + ratio scales have a clear definition of zero. Good examples of ratio variables include height and weight.

Ratio scales provide a wealth of possibilities when it comes to statistical analysis. These variables can be meaningfully added, subtracted, multiplied, divided (ratios). Central tendency can be measured by mode, median, or mean; measures of dispersion, such as standard deviation and coefficient of variation can also be calculated from ratio scales.

Types of variables

	A	B
1	**Temp. in °F**	**Frequency**
2	-15 to 0	2
3	0 to 15	11
4	15 to 30	32
5	30 to 45	61
6	45to 60	87
7	60 to 75	95
8	75 to 90	77
9	**TOTAL**	**365**

Interval

Interval scales are numeric scales where we know both the order of values and the exact differences between them. A classic example is Celsius temperature: the difference between 60° and 50° is the same as between 80° and 70°. Time is another example where increments are equal and measurable.

Interval data allow a wide range of statistical analysis. We can calculate measures of central tendency (mode, median, mean) and measures of dispersion such as standard deviation.

However, interval scales do not have a "true zero" (there is no such thing as "no temperature" on the Celsius scale). Because of this, we can meaningfully add and subtract values, but not multiply or divide to form ratios. For instance, 20° C is not twice as hot as 10° C. Interval scales are therefore very useful, but they do not support ratio comparisons.

Types of variables

Ordinal

Ordinal scales rank values in order, but the exact differences between them are unknown. For example, we know that a rating of 4 is higher than 3 or 2, but we cannot say how much better it is, or whether the gap between "OK" and "Unhappy" equals the gap between "Very happy" and "Happy". Ordinal scales usually measure non-numeric concepts such as satisfaction, happiness or discomfort. Because they only give an order, central tendency should be described using the mode or median; the mean is not appropriate for purely ordinal data.

Example of Ordinal Scales

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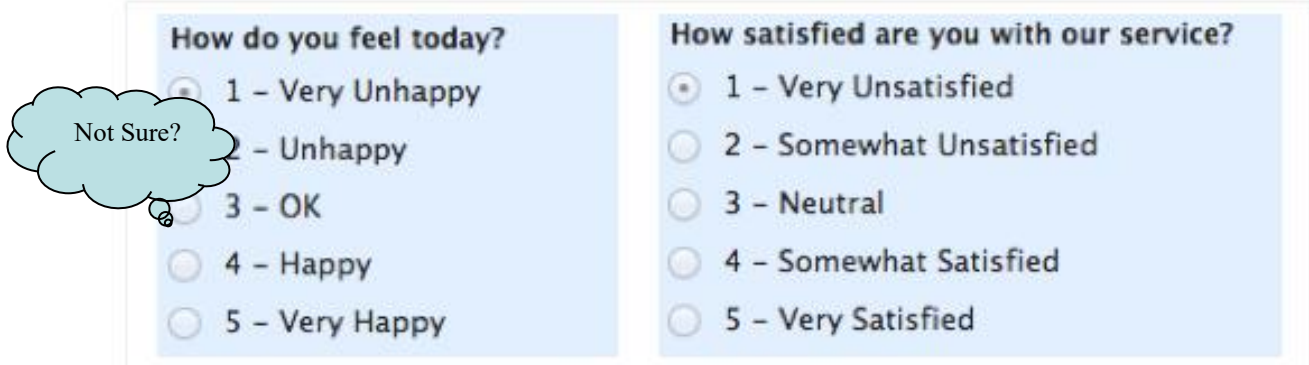
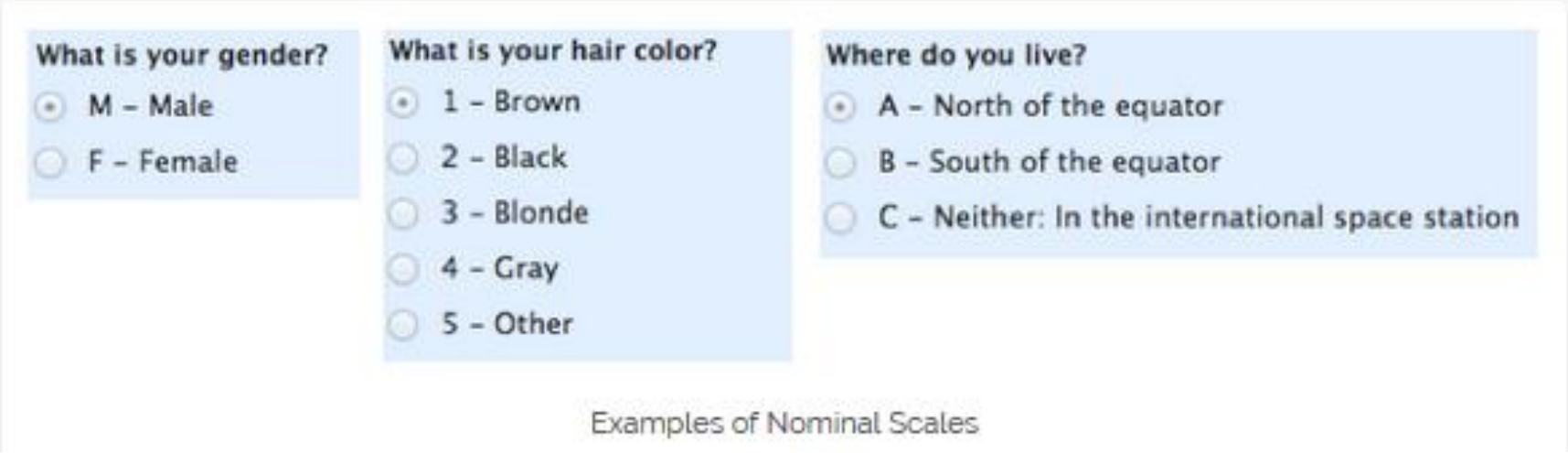


Example of Ordinal Scales

Types of variables

Nominal

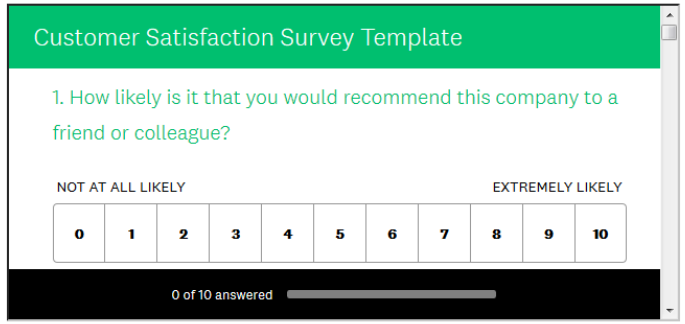
Nominal scales are used only to label categories and have no quantitative meaning. The categories are mutually exclusive, with no ordering or size difference between them. An easy way to remember this is that “nominal” sounds like “name” – these scales are essentially just names or labels for groups.



Examples of Nominal Scales

Survey data:

Asking a number of individuals a set of predefined questions. Using scales to record the answer.



- Each question can be a variable.
- Variables are commonly not single item questions but more complicated constructs.
- There are many **validity** issues including: sampling, inference, questionnaire design, response rates, nonresponse bias, content and convergent validity etc.
- Questionnaire administration involves decisions such as mode (telephone, online, computer or face-to-face), open or closed questions, incentives for completion etc.
- **Always remember:** survey data is about people self-reporting …

Have a think

- Did you ever fill a survey?
- Did you notice it asks the same questions multiple times and in different ways?
- Do you know why?

Reliability and Validity:

Question 1:
Do you agree with this statement? I feel great whenever I enter Primark.
1 – No
2 – Sometimes
3 – All the time
4 – I dream of Primark

Question 2:
How did you feel the last time bought something from Primark?
1- The Best place.
2- Very good.
3- Alright.
4- Not great.

Question 3:
Would you recommend a friend to shop there?
1- Never.
2- Maybe, not sure.
3- Sure.
4- Always!

A person who loves Premark is likely to answer 4,1,4

Those who like it, 3,2,3

Those who are not sure 2,3,2

Those who hate it 1,4,1

Reliability and Validity:

Internal Reliability: use ***Cronbach's Alpha*** *(SPSS) to conduct internal consistency tests on your quantitative method: results have to be higher than 0.7*

Use this if you're data consists of a scale…as in survey data or questionnaire data where participants answer in a scale, like we saw in the earlier example.

Reliability and Validity:

Convergent validity

In the view of some methodologists, the validity of a measure ought to be judged by comparing it to measures of the same concept developed through other methods. For example, if we develop a questionnaire measure of how much time managers spend on various activities (such as attending meetings, touring their organization, informal discussions, and so on), we might examine its validity by tracking a number of managers and using a structured observation schedule to record how much time is spent in various activities and their frequency.

Univariate analysis (one variable)

Univariate analysis is perhaps the simplest form of *statistical analysis*. Like other forms of *statistics*, it *can* be inferential or ***descriptive***. The key fact is that *only* one variable is involved

Univariate analysis involves describing the **distribution of a single variable**, including its **central tendency (including the mean, median, and mode**) and dispersion (including the **range and quartiles** of the data-set, and measures of spread such as the **variance and standard deviatio**n). The shape of the distribution may also be described via indices such as **skewness and kurtosis**. Characteristics of a variable's distribution may also be depicted in graphical or tabular format, including histograms and stem-and-leaf display.

Example of Ratio

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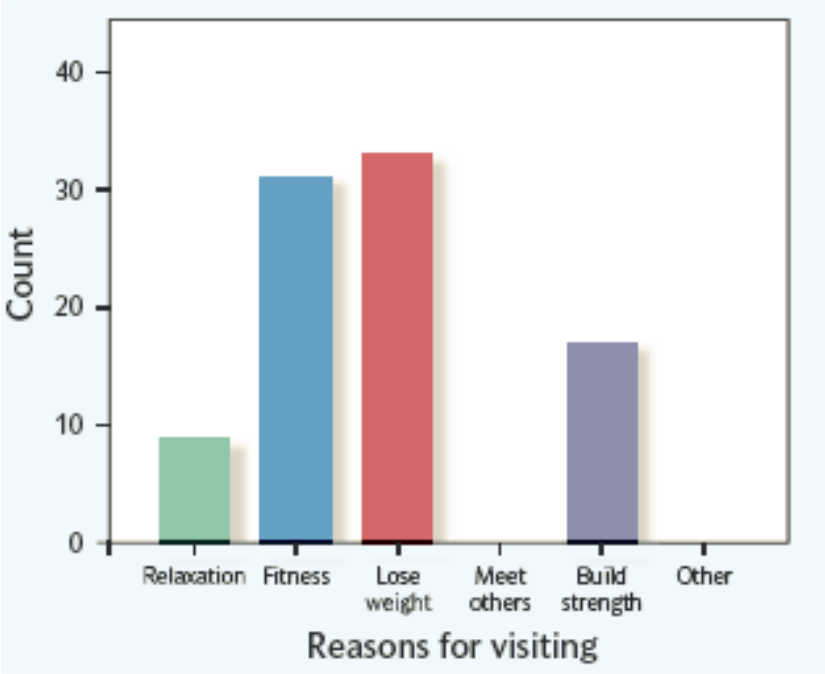
Univariate analysis (one variable)

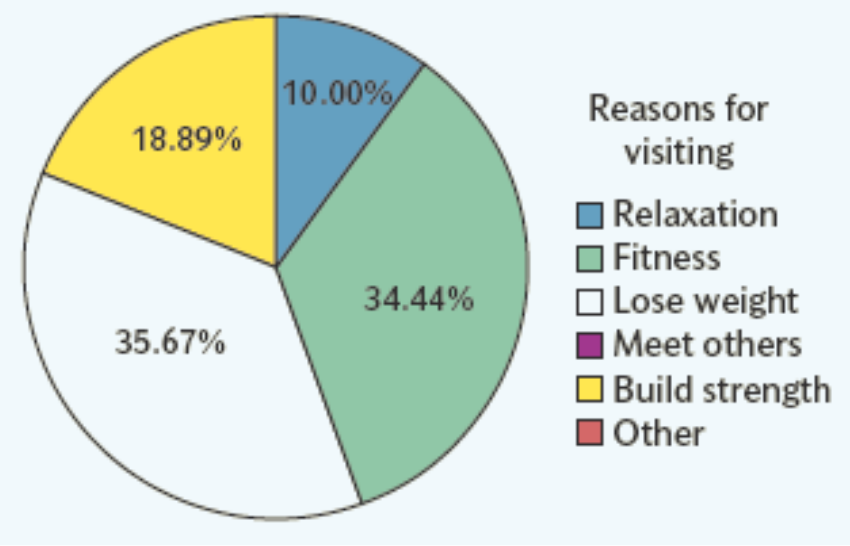
- **Diagrams/charts**
- bar chart or pie chart (nominal or ordinal variables)
- histogram or frequency distribution (interval/ratio variables)

- **Frequency tables**
- number of people or cases in each category
- often expressed as percentages of sample
- interval/ratio data need to be grouped

Visualising one variable: bar and pie charts

Part of designing an App for gym goers you ask:
Main reasons for visiting the gym – bar and pie charts.



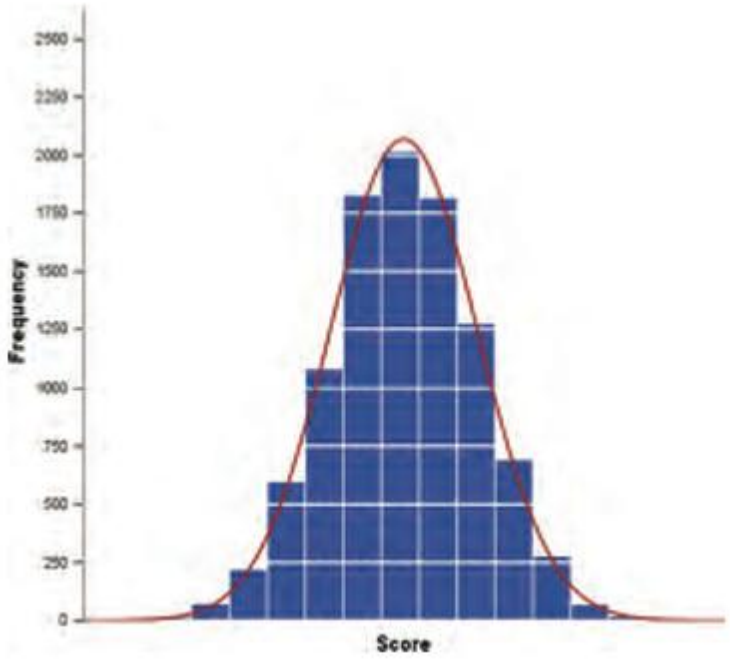


Visualising one variable: frequency distributions

Frequency distribution or histogram is a diagram showing how many times each score of a variable occurs.

Horizontal axis shows the values of the observations

Vertical axis shows the frequency of the observations



Describe a distribution

Shape

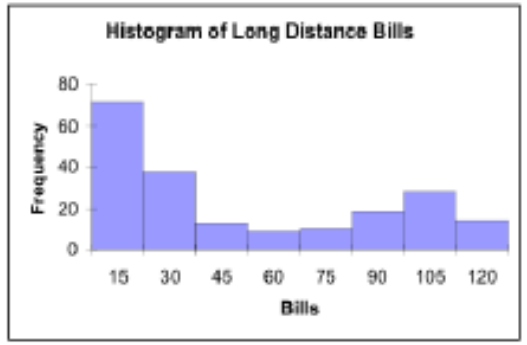
- Humps?
- Symmetry?
- Outliers?

Central tendency

- Mean
- Median
- Mode

Spread

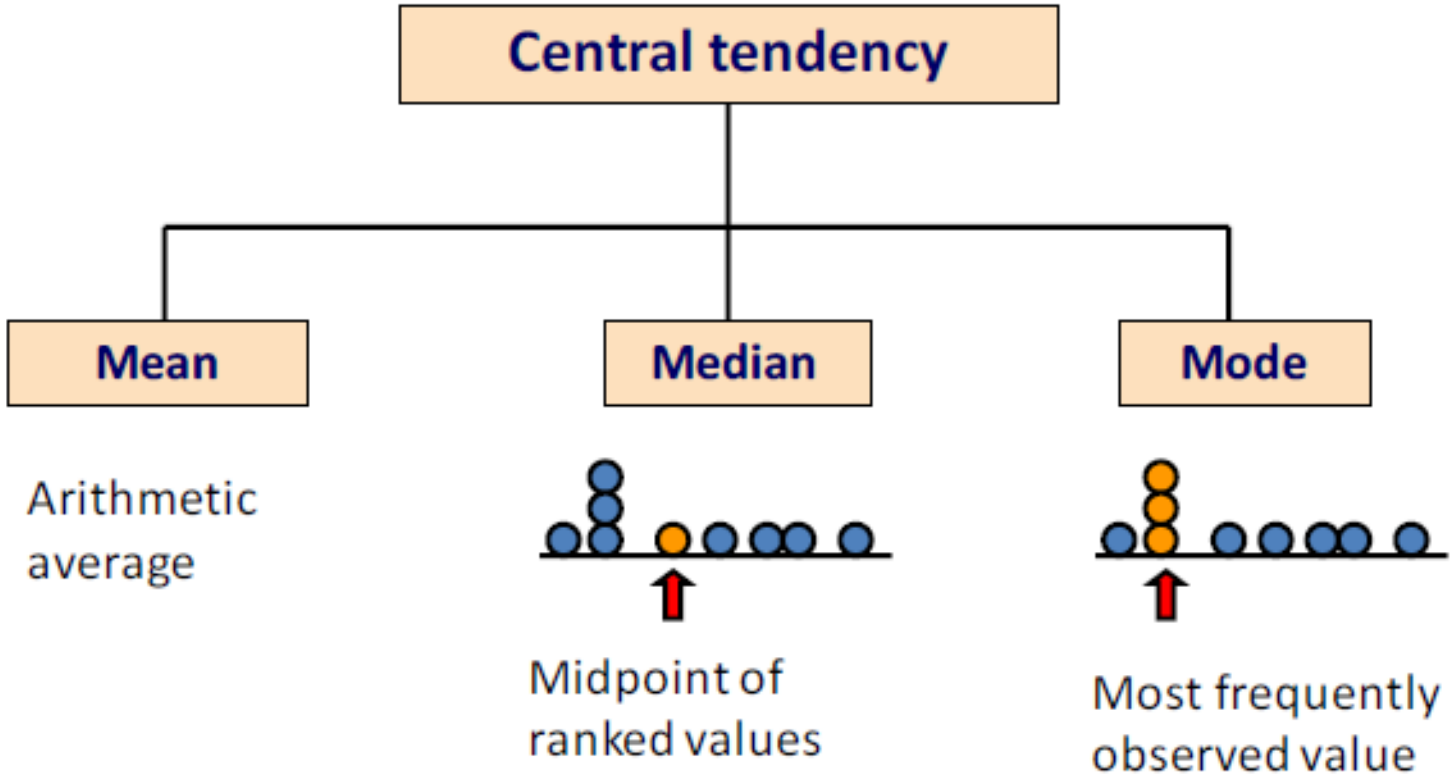
- Range
- Quartiles
- Variance
- Standard deviation
- Coefficient of variation



These measures:

- are never negative
- always depend on the variable's units

Measuring one variable: central tendency



Measuring one variable: range and standard deviation

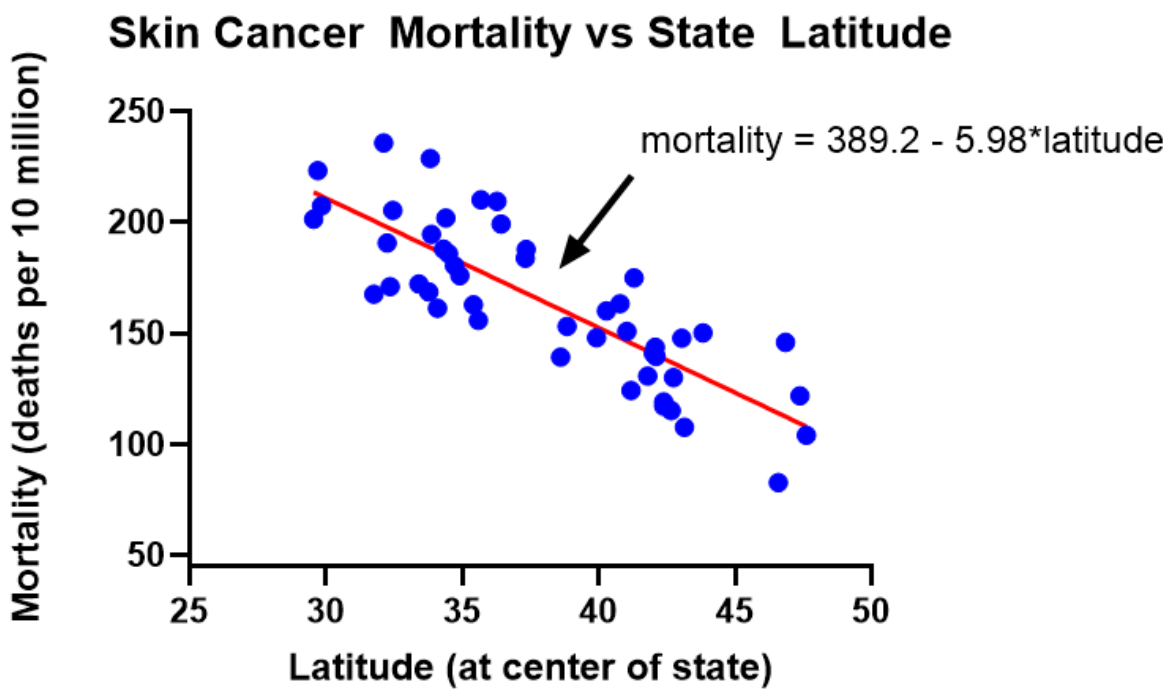
- Measures of dispersion compare levels of variation in different samples to see if there is more variability in a variable in one sample than in another.
- The range is the difference between the minimum and maximum values in a sample.
- The standard deviation is the average amount of variation around the mean, reducing the impact of extreme values (outliers).

Example that explains SD

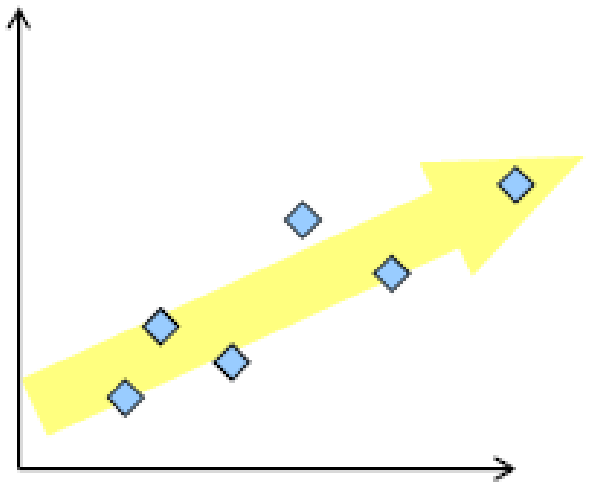
- ----------
- Class average is 60% standard deviation is +/-30%
- Class average is 60% standard deviation is +/-5%
- Class average is 60% standard deviation is +/-10%

Bivariate or multivariate analysis (more than one variable)

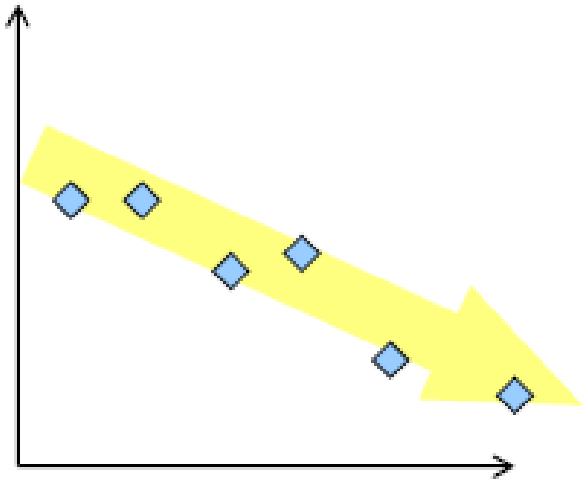
Bivariate analysis means the analysis of bivariate data. It usually involves the variables X and Y. Univariate analysis is the analysis of one (“uni”) variable.



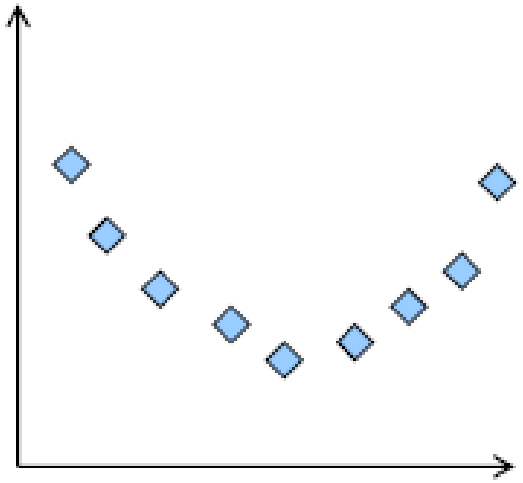
Patterns of scatterplot diagrams



Positive Linear Relationship



Negative Linear Relationship



Nonlinear Relationship

Pearson's r: the relationship between two interval/ratio variables

- Coefficient shows the strength and direction of the relationship, lies between

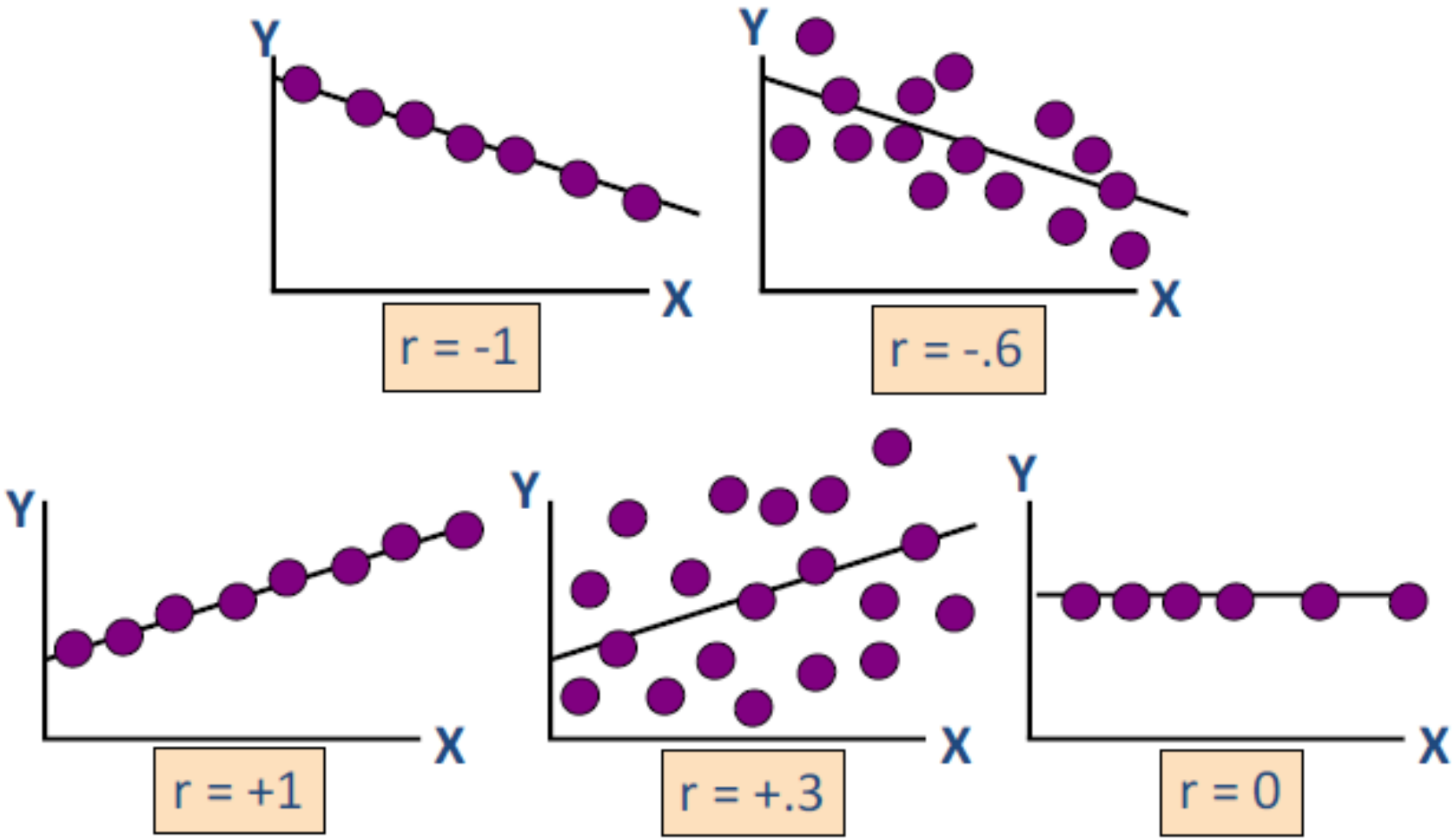
  -1 (perfect negative relationship)

  and

  +1 (perfect positive relationship)

- Relationships must be reasonably linear for the method to work, so remember to plot a scatter diagram first.

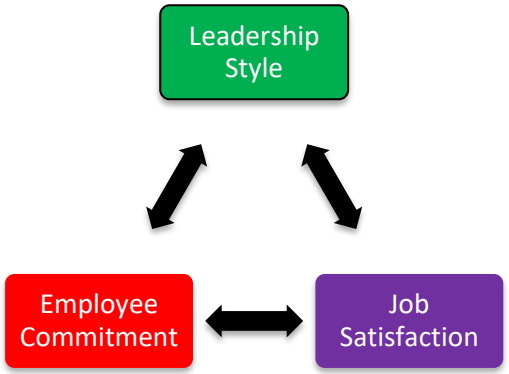
Scatterplots for different correlations



The magnitude is now informative

Correlation is not causation and does not show the direction of the effect

- Whenever we have a strong correlation, it is tempting to explain it by imagining that one variable has caused the other.
- Scatterplots and correlation coefficients never prove causation nor show the direction of the effect.
- Possibly, a hidden variable stands behind a relationship and determines it by simultaneously affecting the other two variables. In this case, it is called a lurking variable.



Employee Commitment ⟷ Job Satisfaction

How do we analyse our quantitative data?

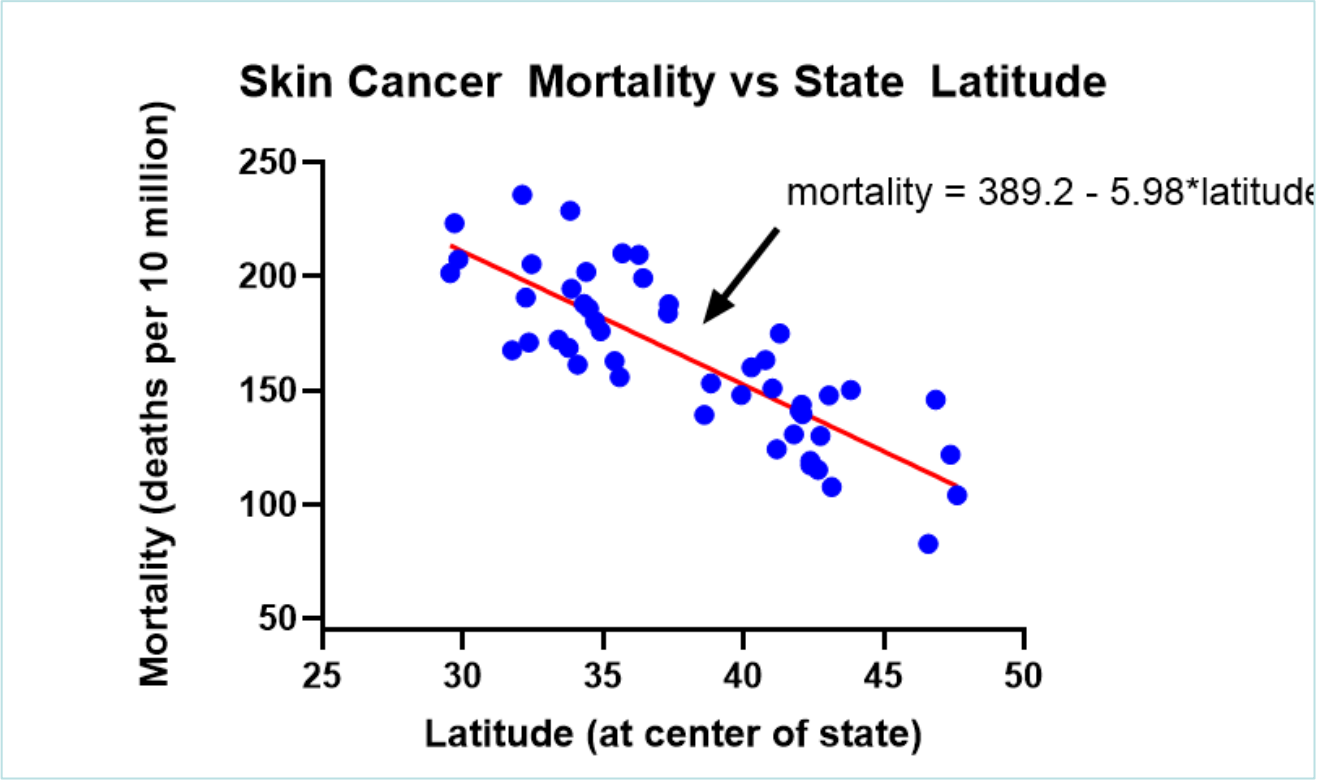
4. What are you looking for?

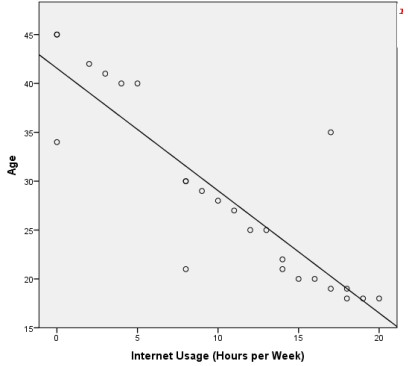
- You can only look for correlation when you have a set of paired data.
- You can look for differences whenever you have two sets of data.
- You can look for prediction when you have - In a prediction study, the goal is to develop a formula for making predictions about the dependent variable, based on the observed values of the independent variables.

Let us examine each of these in more detail:

Bivariate or multivariate analysis (more than one variable)

Bivariate analysis means the analysis of bivariate data. It usually involves the variables X and Y. Univariate analysis is the analysis of one ("uni") variable.





1. Tests of Correlation:

Pearson correlation

Tests for the strength of the association between two continuous (***ratio* or *interval***) variables.

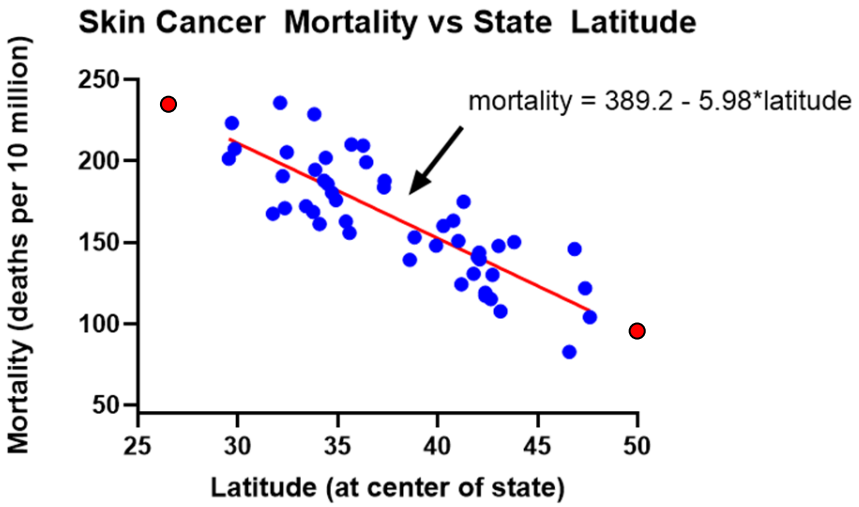
Spearman correlation

Tests for the strength of the association between two ***ordinal*** variables (does not rely on the assumption of normal distributed data)

Chi-square

Tests for the strength of the association between two ***nominal*** variables

2. Tests of Prediction:



<u>Simple linear regression</u>

Tests how a change in the predictor variable predicts the level of change in the outcome variable. In **simple linear regression** a single independent variable is **used** to predict the value of a dependent variable.

<u>Multiple regression</u>

Tests how a change in the combination of two or more predictor variables predict the level of change in the outcome variable. In multiple **linear regression,** two or more independent variables are **used** to predict the value of a dependent variable. The difference between the two is the number of independent variables.

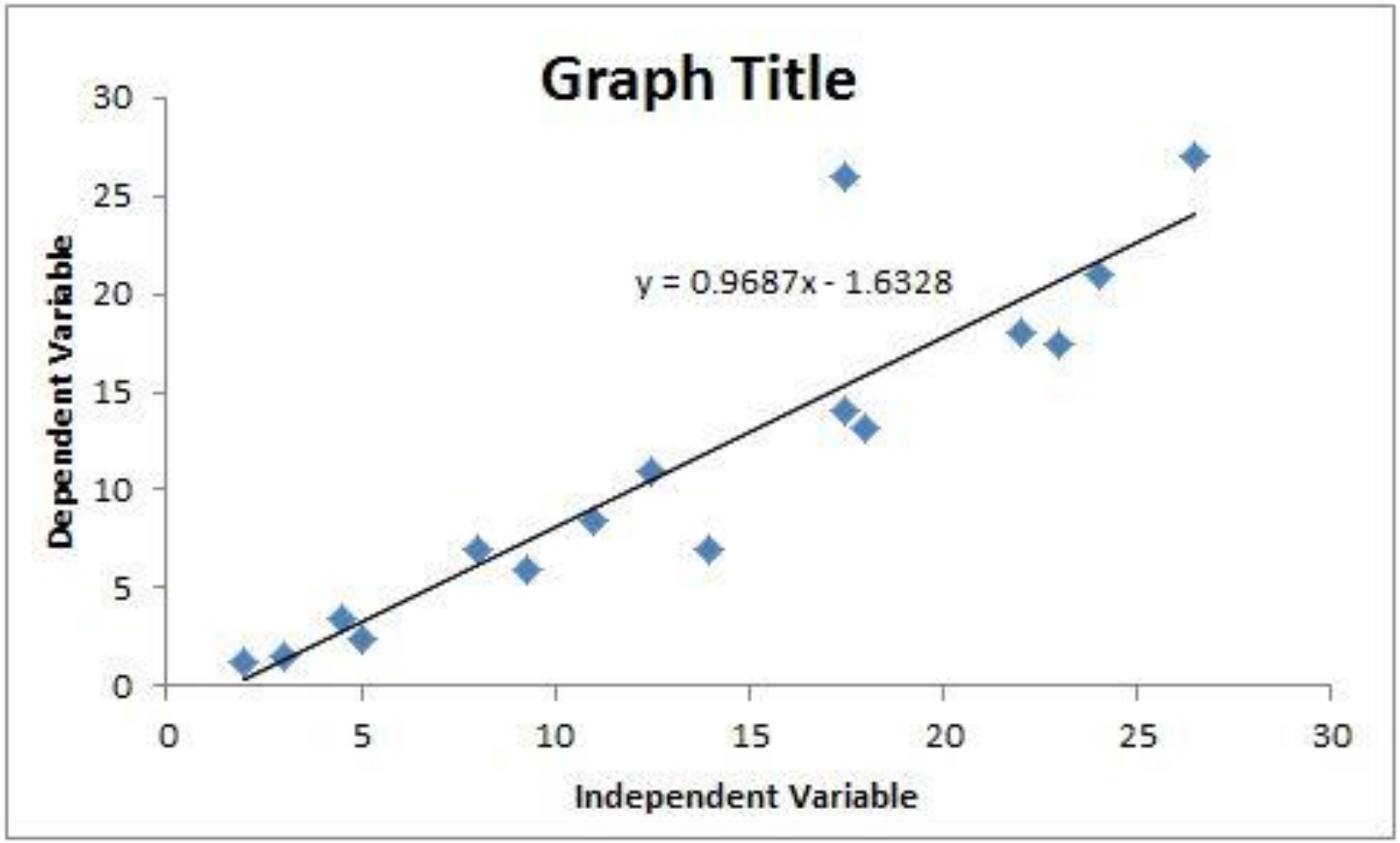
Variables:

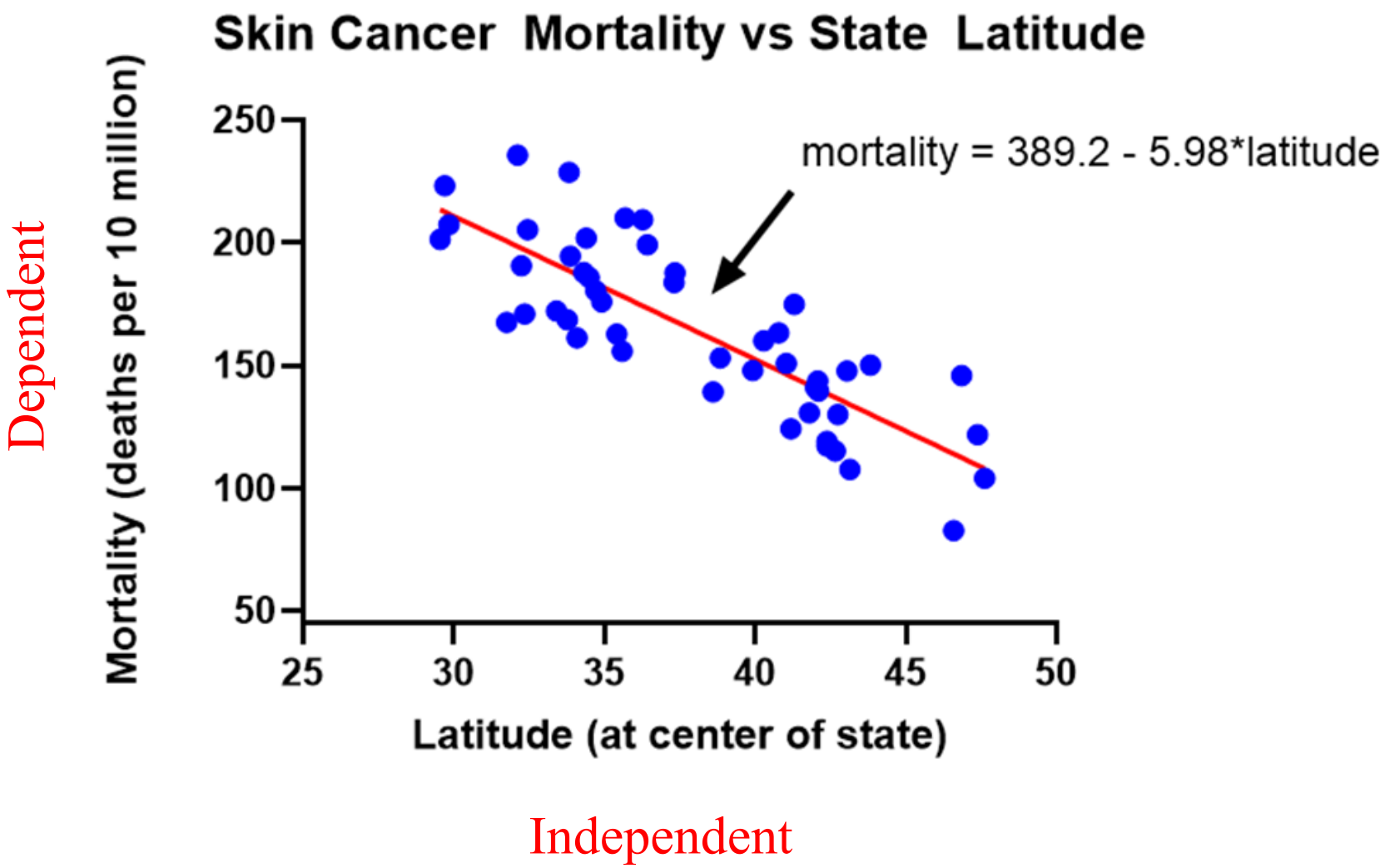
The two main variables in an experiment are the independent and dependent variable.

An **independent variable** is the variable that is changed or controlled in a scientific experiment to test the effects on the dependent variable.

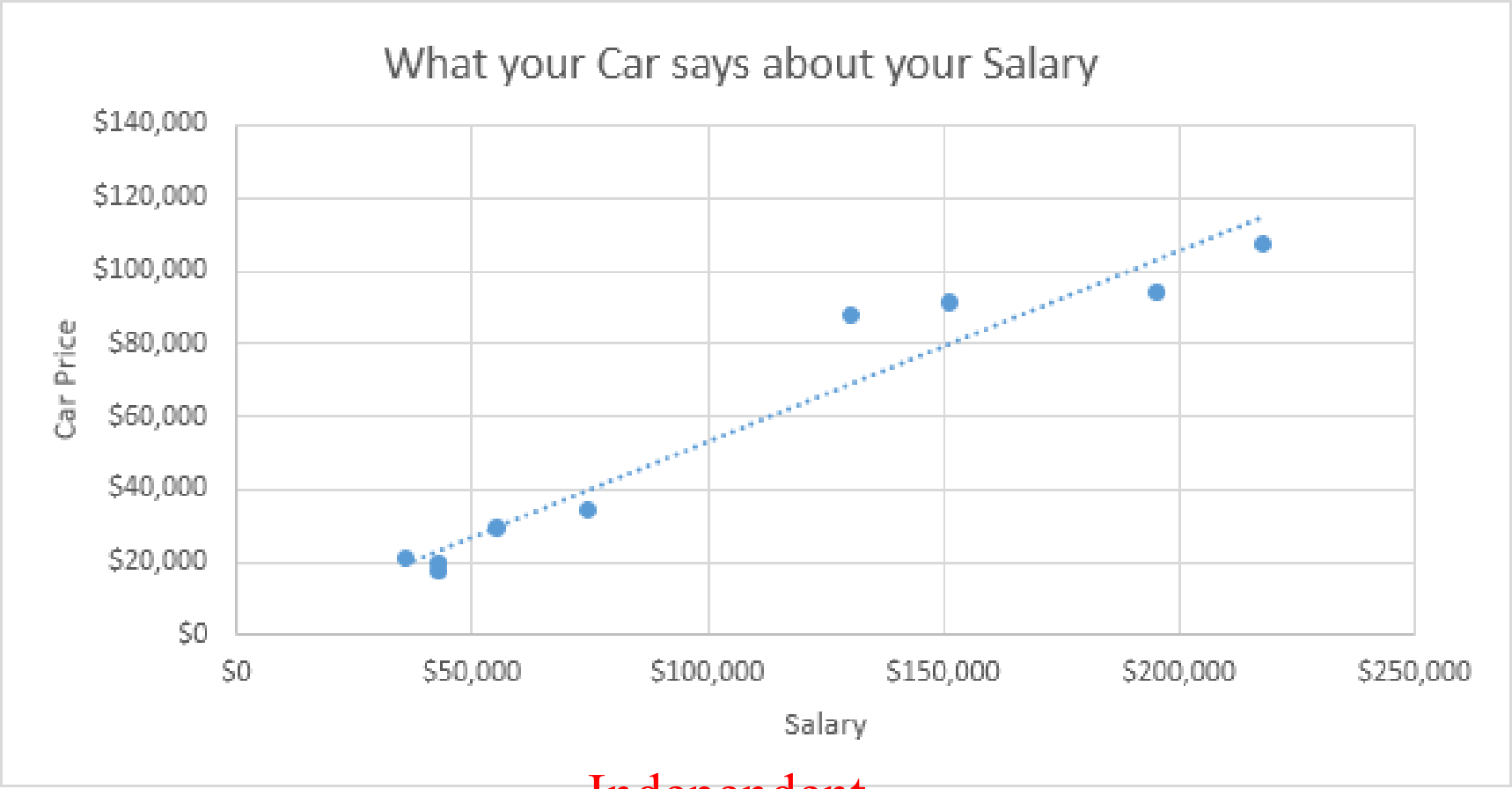
A **dependent variable** is the variable being tested and measured in a scientific experiment.
The dependent variable is 'dependent' on the independent variable. As the experimenter changes the independent variable, the effect on the dependent variable is observed and recorded.

Let us see few examples:

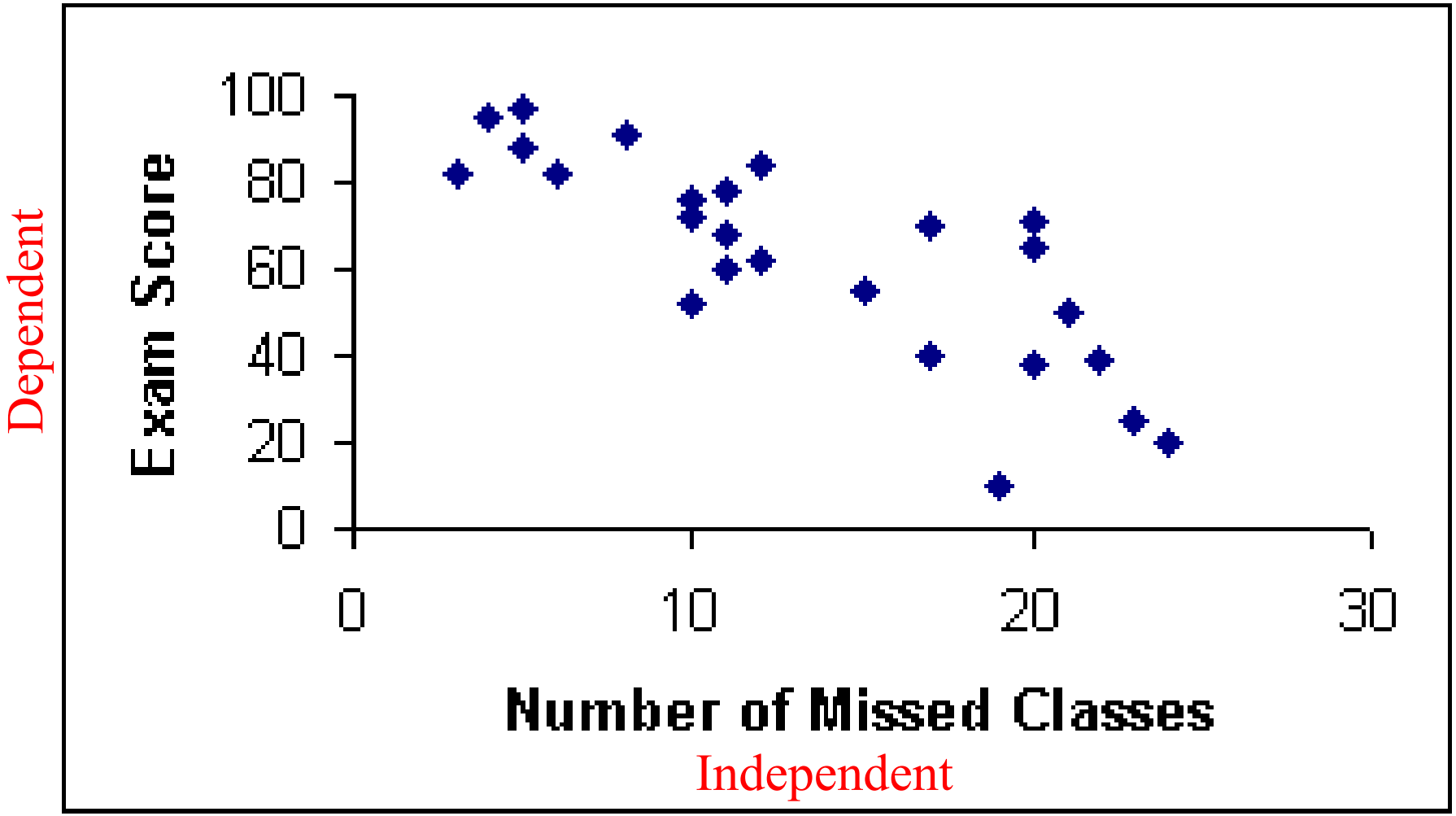

Graph Title
y = 0.9687x - 1.6328
Dependent Variable
Independent Variable
0
5
10
15
20
25
30


Skin Cancer Mortality vs State Latitude
mortality = 389.2 - 5.98*latitude
Mortality (deaths per 10 million)
250
200
150
100
50
25
30
35
40
45
50
Latitude (at center of state)
Dependent
Independent

Dependent



Independent


Dependent
Exam Score
100
80
60
40
20
0
0
10
20
30
Number of Missed Classes
Independent

Have a think:

- Consider the benefit of having one of these tools in your study?
  - Correlation
  - Comparison of Means
  - Prediction

Internet survey systems

Web survey systems provide advanced support with features such as predefined questions, custom collectors, crosstabulation reporting etc.

SurveyShare

SurveyMonkey

Google Docs

QuestionPro
Online Research Made Easy™

LimeSurvey

qualtrics.com

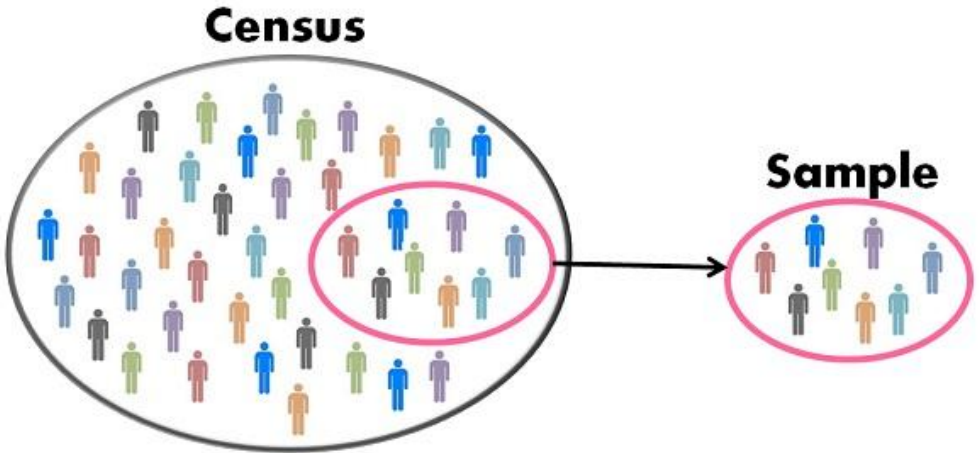
Strongly recommended for UEL

Microsoft Forms

What is Sampling?

- **Sampling**: Sampling is the process of selecting something part of something larger.

- **Population** (also called universe) represents the whole from which we select a sample from.
- There are several examples of sampling:
  - Sampling people for a medical experiment
  - Sampling products before you buy
  - Sampling people's opinion before elections



Why do we sample?

- But Why do we sample?
  - Why don't we experiment on everyone, try all products,
  - or ask everyone's opinion before an election?

- Cost!!!
  - It would take a lot of time to test a medical discovery on everyone.
  - It would cost a lot to try all the products before buying them.
  - It would be impossible to ask everyone's opinion before an election.

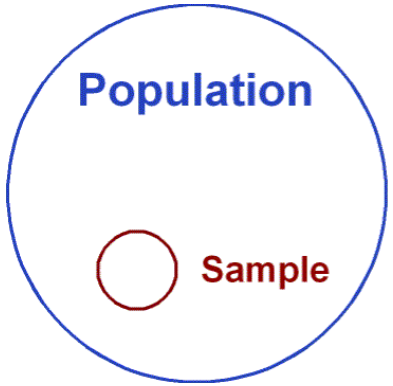
Sample Selection:

A subset of the population.

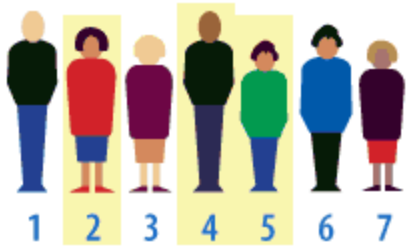
- We have two types of sampling:
  - **<u>Probability Sampling</u>**: This involves procedure whereby each element in the population has equal chance of being selected.
  - **<u>Non-probability Sampling:</u>** This does not involve scientific procedure.
- The main difference being, with probability sampling we can estimate/calculate our possible sample error,
- Whereas in the non-probability, we cannot! And therefore, you cannot generalise!

Types of samples taking:

- **Non-probability methods:**
  - Convenience Sampling:
    - Sample members are chosen on the basis of their being readily available.
  - Judgmental Sampling:
    - Members are screened to see if they suit this sampling, but the method of selection cannot be scientifically generalized on.
  - Quota sampling:
    - Members are selected by convenience but with specific groups quota. Insuring you have 50% males/ 50% females…etc.



Types of samples taking:



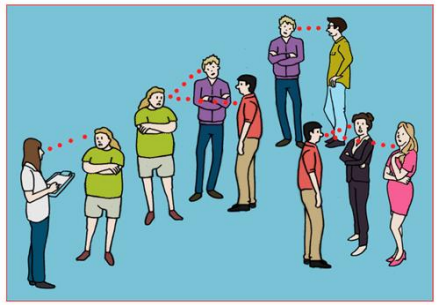
Assign Numbers, Auto-Generate Random Selections

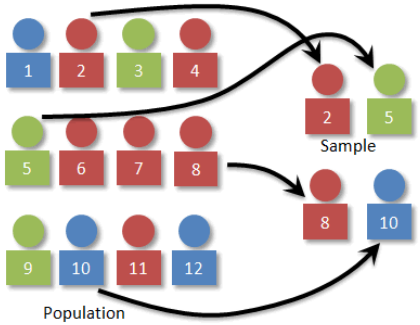
- Most Common **Probability methods**:
  1. *Simple Random Sampling:*
     - Sample members are selected randomly with each member of the population having equal chance of being selected.
  2. *Systematic Sampling:*
     - Sample members are selected at regular intervals after a random number start. Each interval is called an N.

1	26	51	76
2	27	52	77
3	28	53	78
4	29	54	79
5	30	55	80
6	31	56	81
7	32	57	82
8	33	58	83
9	34	59	84
10	35	60	85
11	36	61	86
12	37	62	87
13	38	63	88
14	39	64	89
15	40	65	90
16	41	66	91
17	42	67	92
18	43	68	93
19	44	69	94
20	45	70	95
21	46	71	96
22	47	72	97
23	48	73	98
24	49	74	99
25	50	75	100

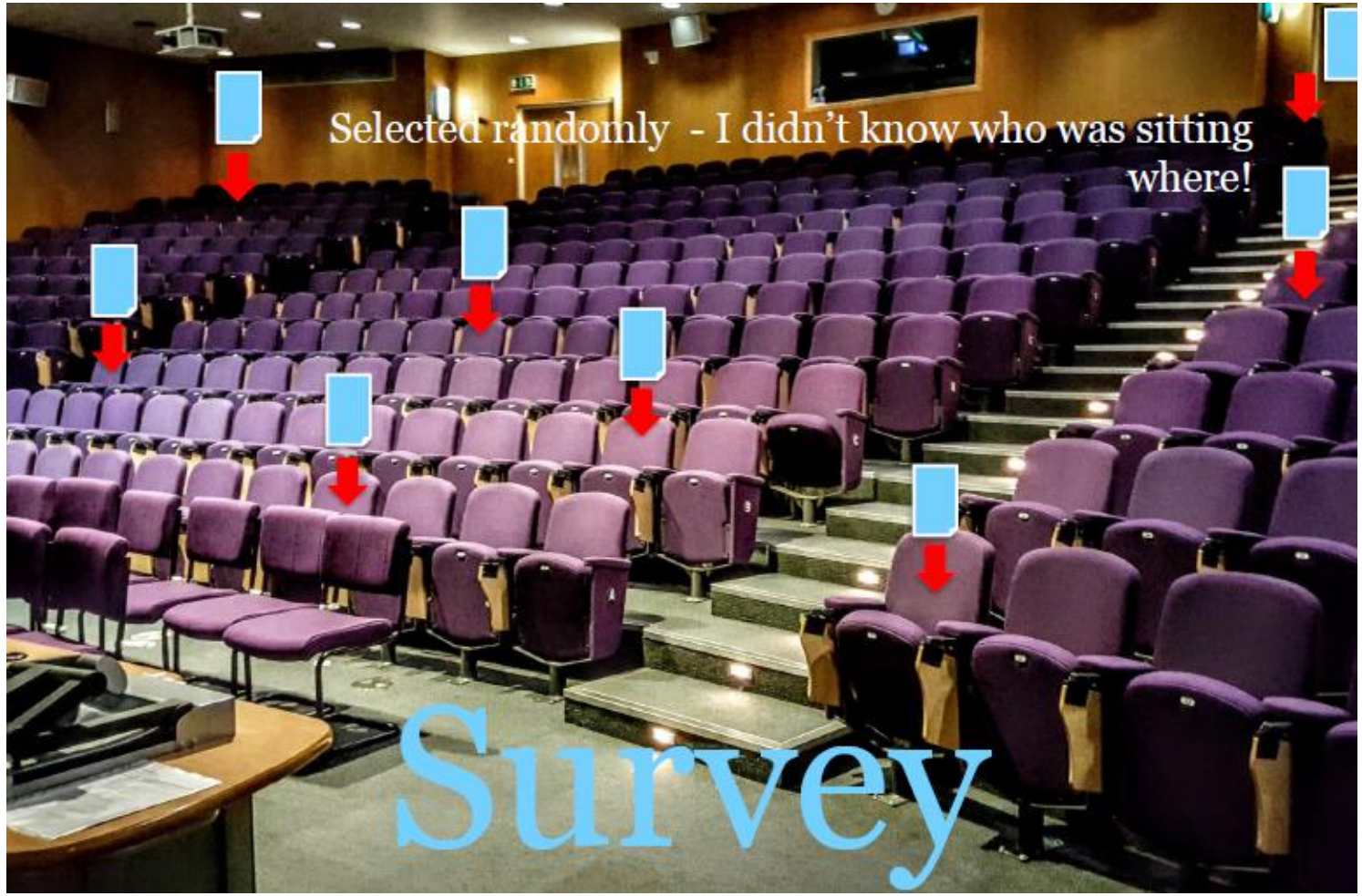
Types of samples taking:

- **Probability methods:**
  - *3. Multiplicity (snowball) Sampling:*
    - Sample members are initially chosen by any probability sampling method and are subsequently asked to identify others with the desired characteristics.
  - *4. Stratified Sampling:*
    - If you have different segments (categories) in the population and you want to insure that after your random selection is done, you have equal representations.





Simple random



Systemic random

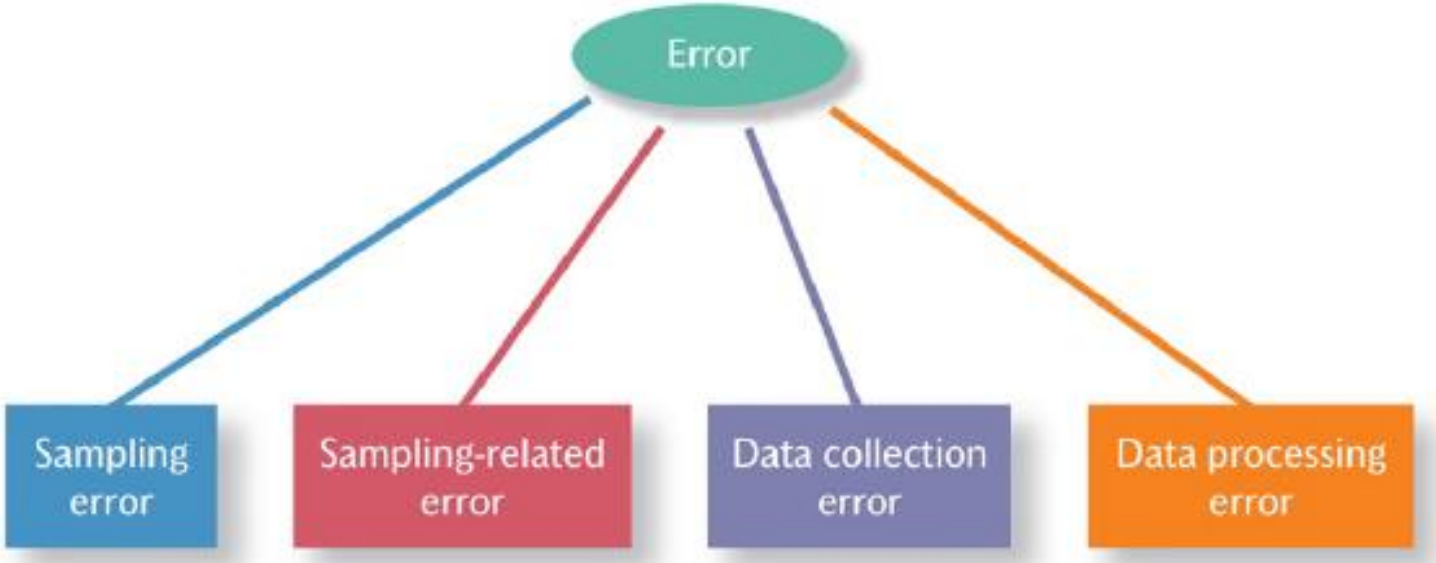
Stratified random

Strata 1: Male
Strata 2: Female

Have a think?

- What is the point of doing a ‘non-probability’ sampling if you cannot generalise in the data?

Sources of error in survey research



Sampling error	Sampling-related error	Data collection error	Data processing error
Selection of sample and generalisability	Sampling process	Questionnaire design and administration	Errors in coding and analysing answers

Further important issues about surveys …

- **Legal and ethical issues:** informed consent, privacy issues and how the results of the study will be used.

- **Self-reporting bias** is inevitable to some extent
  - People do not remember, misinterpret questions or simply do not want to report on certain things especially to an unknown researcher!

- **Replication:** although it is almost impossible to replicate a survey with the same results, replication should be ensured by documenting the process and choices made.

- **Post-survey adjustments involve issues such as:**
  - Data cleaning and defining number of usable responses
  - Treating missing values within respondents
  - Content analysis of open-ended questions

Any Questions?